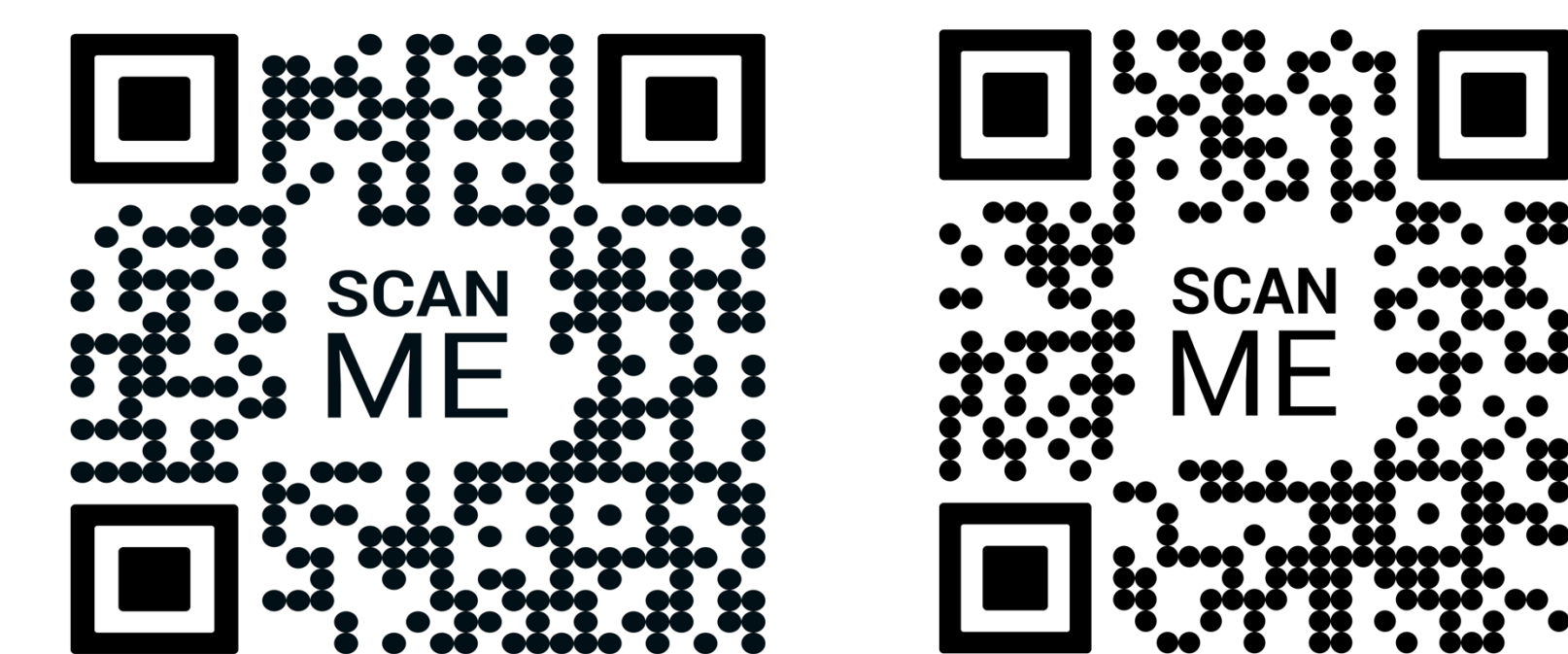
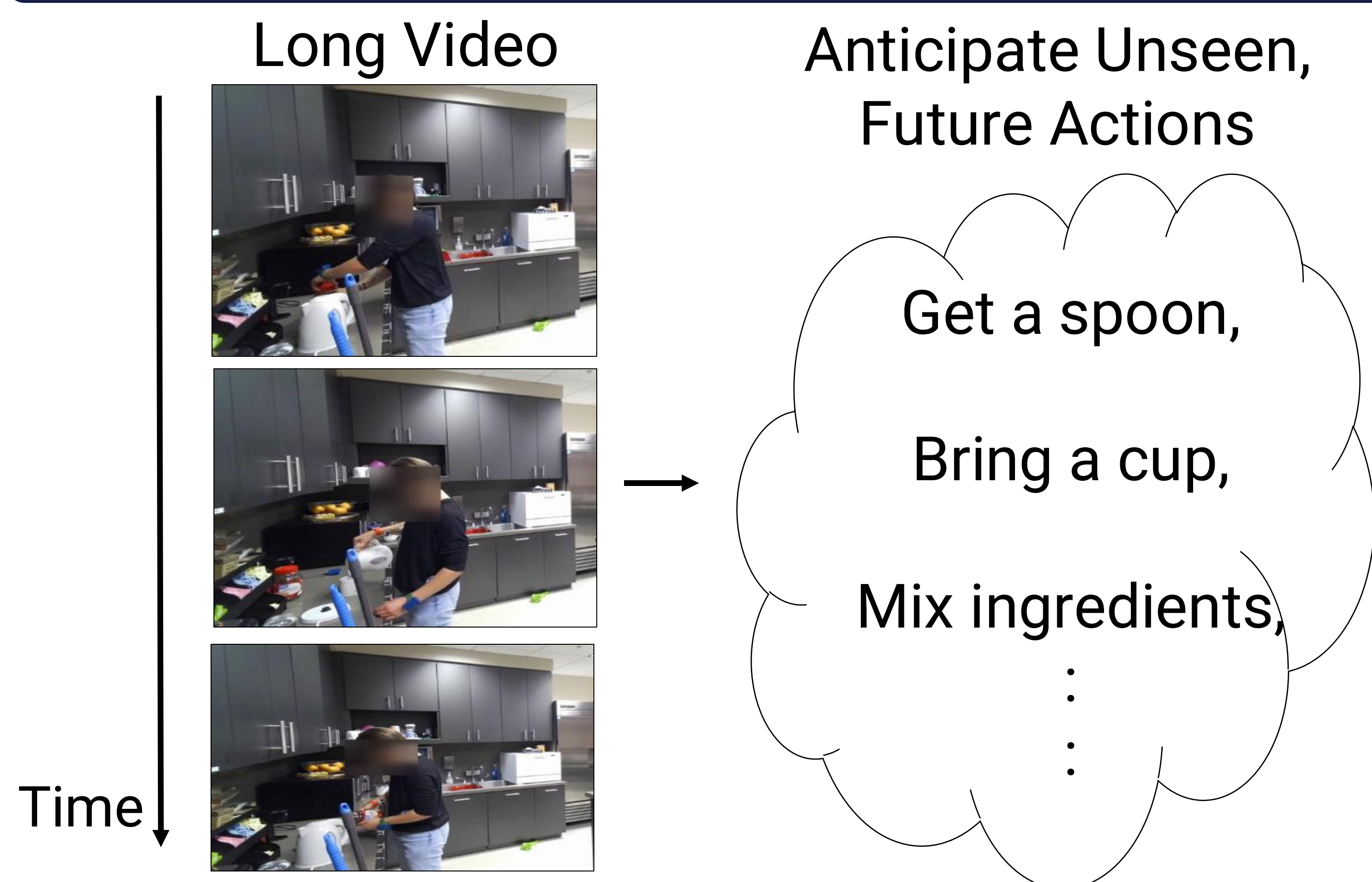




## Contribution

- ✓ Multi-level (Hierarchical) & Multi-modal Modeling
- ✓ Fine-grained Action Text Generator
- ✓ Benchmark on our newly released *DARai Dataset*

## What is Action Anticipation?



## Challenge of Action Anticipation

### 1. Visual information (RGB) alone lacks clarity



What is the person doing in the kitchen?

### 2. Ambiguity leads to multiple plausible futures



Wash dishes?  
Add flour?  
Get a cup?

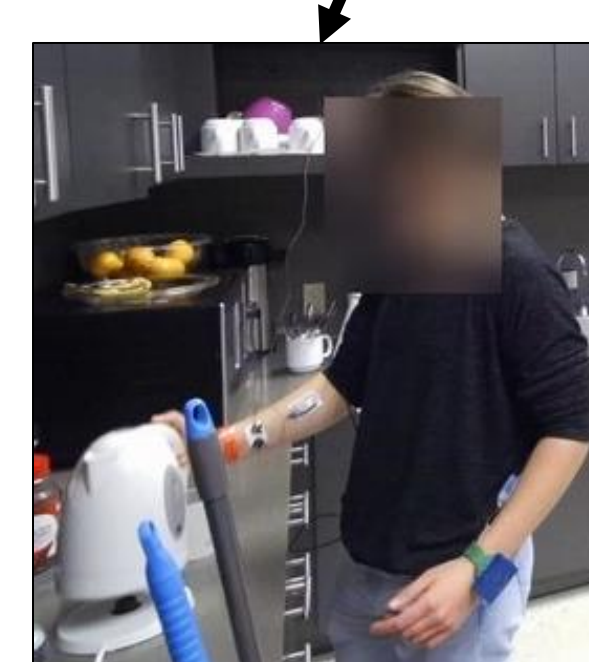
Ambiguous Video

Multiple Plausible futures

### 3. Ambiguity increases uncertainty of future actions

## Idea

### 1. Multi-level: Hierarchical Decomposition of Video



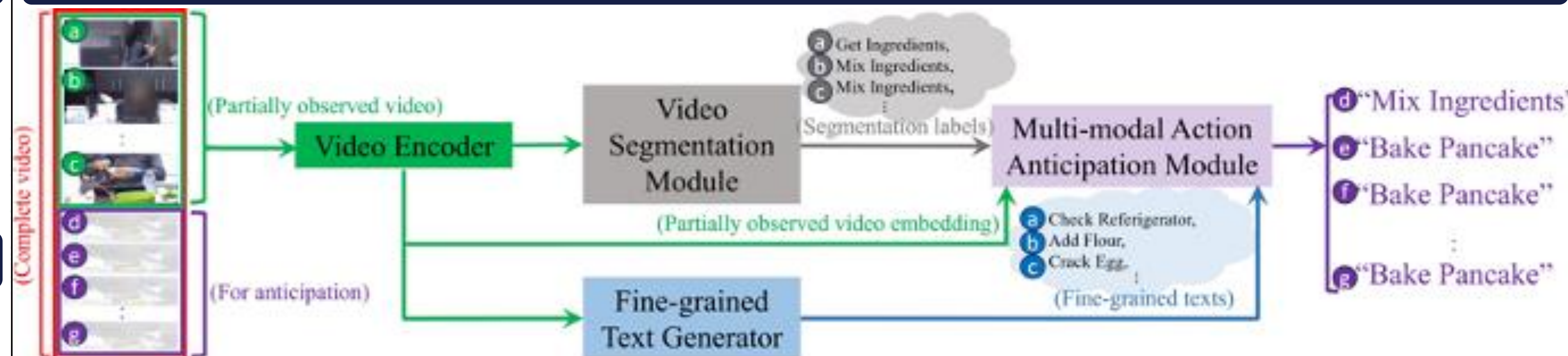
"Pour water"

"Add coffee"

"Pour water"

### 2. Multi-modal: Generate fine-grained texts

## Architecture



### 1). Video Segmentation Module

- Transforms each video frame into action labels.

### 2). Fine-grained Text Generator

- Transforms each video frame into fine-grained texts.

### 3). Multi-modal Action Anticipation Module

- Aligns **RGB** with **Text** modality and anticipate next actions.

## Fine-grained text generator

### Hierarchical Decomposition of Video



t=1-8, Mix Ingredients

- $x_1$  t=1, Pour water
- $x_2$  t=2, Pour water
- $x_3$  t=3, Pour water
- $x_4$  t=4, Add coffee
- $x_5$  t=5, Add coffee
- $x_6$  t=6, Pour water
- $x_7$  t=7, Pour water
- $x_8$  t=8, Pour water

$$Loss = L_{Cross\ Entropy} + \alpha L_{intra} + \beta L_{inter}$$

### 1. $L_{intra}$ : Pull Frames within the same temporal interval

$$L_{intra} = \sum_k \sum_{x_i \in X(k)} \|x_i - \mu_k\|^2, \quad \mu_k = \frac{1}{X(k)} \sum_{x_i \in X(k)} x_i$$

### 2. $L_{inter}$ : Push Frames that are temporally distant

$$L_{inter} = \sum_{k, k' \neq k} \frac{1}{\|\mu_k - \mu_{k'}\|}$$

## Results (Evaluation Metric: Mean of Classes)

| Dataset   | Methods | $\alpha = 0.2$ |              |              |              | $\alpha = 0.3$ |              |              |              |
|-----------|---------|----------------|--------------|--------------|--------------|----------------|--------------|--------------|--------------|
|           |         | $\beta(0.1)$   | $\beta(0.2)$ | $\beta(0.3)$ | $\beta(0.5)$ | $\beta(0.1)$   | $\beta(0.2)$ | $\beta(0.3)$ | $\beta(0.5)$ |
| Breakfast | RNN     | 18.11          | 17.20        | 15.94        | 15.81        | 21.64          | 20.02        | 19.73        | 19.21        |
|           | CNN     | 17.90          | 16.35        | 15.37        | 14.54        | 22.44          | 20.12        | 19.69        | 18.76        |
|           | FUTR    | 47.06          | 47.08        | 47.11        | 47.11        | 65.55          | 65.55        | 65.54        | 65.51        |
|           | GTD     | 49.86          | 49.75        | 49.65        | 49.58        | 65.00          | 64.39        | 63.95        | 63.79        |
|           | Ours    | 50.55          | 50.53        | 50.52        | 50.56        | 66.49          | 66.53        | 66.63        | 66.68        |
| 50Salads  | RNN     | 30.06          | 25.43        | 18.74        | 13.49        | 30.77          | 17.19        | 14.79        | 09.77        |
|           | CNN     | 21.24          | 19.03        | 15.98        | 09.87        | 29.14          | 20.14        | 17.46        | 10.86        |
|           | FUTR    | 54.83          | 54.33        | 52.45        | 51.03        | 100.00         | 100.00       | 98.62        | 53.00        |
|           | Ours    | 69.61          | 69.27        | 69.73        | 48.35        | 100.00         | 99.88        | 98.70        | 39.61        |

| $\alpha$ | Methods | $\beta(0.1)$ | $\beta(0.2)$ | $\beta(0.3)$ | $\beta(0.5)$ |
|----------|---------|--------------|--------------|--------------|--------------|
| 0.1      | FUTR    | 24.26        | 24.25        | 24.74        | 23.46        |
|          | AFFT    | 20.25        | 23.13        | 23.63        | 23.42        |
|          | Ours    | 26.53        | 25.82        | 25.98        | 24.71        |
| 0.2      | FUTR    | 25.05        | 25.11        | 24.48        | 23.18        |
|          | AFFT    | 23.14        | 24.78        | 23.62        | 21.02        |
|          | Ours    | 25.75        | 25.70        | 24.24        | 23.12        |
| 0.3      | FUTR    | 40.71        | 33.57        | 33.42        | 30.79        |
|          | AFFT    | 33.82        | 29.25        | 28.33        | 25.45        |
|          | Ours    | 42.00        | 34.71        | 34.49        | 31.34        |

## ▲ Comparison with the State-of-the-arts across 3 Different Human Activity Datasets

| Dataset   | Methods                  | Observation Rate ( $\alpha$ ) |       |       |       |
|-----------|--------------------------|-------------------------------|-------|-------|-------|
|           |                          | 0.1                           | 0.2   | 0.3   | 0.4   |
| Breakfast | w/o multi-level modeling | 25.51                         | 47.10 | 65.54 | 68.24 |
|           | w/ multi-level modeling  | 30.05                         | 50.54 | 66.59 | 67.20 |
| 50Salads  | w/o multi-level modeling | 31.25                         | 53.16 | 87.91 | 68.80 |
|           | w/ multi-level modeling  | 39.33                         | 64.24 | 84.55 | 79.82 |
| DARai     | w/o multi-level modeling | 24.18                         | 24.45 | 34.63 | 34.16 |
|           | w/ multi-level modeling  | 25.76                         | 24.70 | 35.64 | 38.36 |

## ▲ Hierarchical (Multi-level) Modeling

## ▲ Multi-modal Modeling

$\alpha$ : Observation Rate  
 $\beta$ : Prediction Rate

| Dataset   | Methods     | Observation Rate (Input) |       |       |       |
|-----------|-------------|--------------------------|-------|-------|-------|
|           |             | 0.1                      | 0.2   | 0.3   | 0.4   |
| Breakfast | Uni-modal   | 26.86                    | 47.21 | 62.43 | 66.64 |
|           | Multi-modal | 30.05                    | 50.54 | 66.59 | 67.20 |
| 50Salads  | Uni-modal   | 29.76                    | 53.76 | 90.92 | 73.92 |
|           | Multi-modal | 39.33                    | 64.24 | 84.55 | 79.82 |
| DARai     | Uni-modal   | 25.87                    | 22.02 | 28.92 | 27.08 |
|           | Multi-modal | 25.76                    | 24.70 | 35.64 | 38.36 |